Assignment 2

Chul Hee Kim

2021 9 17

Problem 1

```
if(!require(tidyverse)) install.packages("tidyverse", repos = "http://cran.us.r-project.org")
if(!require(caret)) install.packages("caret", repos = "http://cran.us.r-project.org")
library(tidyverse)
library(caret)
crime_data <- read.csv("/cloud/project/Data/Crime Rate.csv")
```

a-)

```
fit_crime <- lm(Y ~ X, data = crime_data)
anova(fit_crime)
```

```
## Analysis of Variance Table
##
## Response: Y
##           Df    Sum Sq  Mean Sq F value    Pr(>F)
## X          1  93462942 93462942  16.834 9.571e-05 ***
## Residuals 82 455273165  5552112
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

b-)

```
n <- nrow(crime_data)
f <- qf(0.95, 1, n-2)
```

$H_0 : \beta_1 = 0$
$H_a : \beta_1 \neq 0$

ANOVA table above shows $F^* = 16.834$, and $F(1-\alpha; 1, n-2) = 3.957$. Since F-statistic is bigger, we conclude H_a, and the model is significant.

```
summary(fit_crime)
```

```
##
## Call:
## lm(formula = Y ~ X, data = crime_data)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -5278.3 -1757.5  -210.5  1575.3  6803.3
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
```

```
## (Intercept) 20517.60    3277.64   6.260 1.67e-08 ***
## X            -170.58      41.57  -4.103 9.57e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2356 on 82 degrees of freedom
## Multiple R-squared:  0.1703, Adjusted R-squared:  0.1602
## F-statistic: 16.83 on 1 and 82 DF,  p-value: 9.571e-05
```

```
t <- qt(0.975,n-2)
```

The summary above shows that the t^* for X is -4.103. The equivalence of F test and t test says that $F^* = (t^*)^2$, and it is consistent with above results since

$(t^*)^2 = (-4.103)^2 = 16.83 = F^*$.

$F(1-\alpha; 1, n-2) = 3.957$
$[t(1-\alpha/2; n-2)]^2 = 3.957$

Therefore, we have shown the numerical equivalence of the two test statistics and decision rules, which should be the case with one-predictor-variable model. The p-values for the two tests are also shown in the summary above. The one for t test appears in the row same as X and reads 9.57e-05. The one for F test appears at the bottom and reads 9.571e-05, and those two p-values are the same.

c-) As the R-squared value in the summary shown in (b) explains, the total variation in crime rate is reduced by 17.03% when the percentage of high school graduates is introduced into the analysis. This is a relatively small reduction because there is a lot of room for more reduction by introducing other variables. However, one can still argue that the percentage of high school graduates is significant, given its small p-value. The point we want to emphasize here is that the model still falls short of predicting the crime rate.

d-)

```
b0_full <- fit_crime$coefficients[1]
b1_full <- fit_crime$coefficients[2]

fit_crime_red <- lm(Y ~ 1, data = crime_data)
b0_red <- fit_crime_red$coefficients[1]
```

Full Model: $Y_i = \beta_0 + \beta_1 X_i + \epsilon_i = 2.05176 \times 10^4$ + -170.575X + ϵ_i

Reduced Model: $Y_i = \beta_0 + \epsilon_i = 7111.202 + \epsilon_i$

e-)

```
anova(fit_crime)
```

```
## Analysis of Variance Table
##
## Response: Y
##           Df    Sum Sq  Mean Sq F value    Pr(>F)
## X          1  93462942 93462942  16.834 9.571e-05 ***
## Residuals 82 455273165  5552112
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
anova(fit_crime_red)
```

```
## Analysis of Variance Table
##
## Response: Y
```

```
##           Df    Sum Sq Mean Sq F value Pr(>F)
## Residuals 83 548736108 6611278
```

```
SSE_F <- anova(fit_crime)$`Sum Sq`[2]
SSE_R <- anova(fit_crime_red)$`Sum Sq`[1]
f_stat <- ((SSE_R-SSE_F)/(83-82)) / (SSE_F/82)
f_gl <- qf(0.95,1,82)
```

(1) SSE(F) $= 4.5527317 \times 10^8$
(2) SSE(R) $= 5.4873611 \times 10^8$
(3) df_F = n - 2 = 82
(4) df_R = n - 1 = 83
(5) $F^* = \frac{SSE(R)-SSE(F)}{df_R - df_F} \div \frac{SSE(F)}{df_F} = \frac{455{,}273{,}165-548{,}736{,}108}{83-82} \div \frac{548{,}736{,}108}{82} = 16.834$
(6) $F(1-\alpha; df_R - df_F, df_f) = F(0.95; 1, 82) = 3.957$

f-) Yes, the F^* and the decision rule for the general linear test numerically equivalent to those in part a,b as the results above show.

Problem 2

```
X <- c(4,8,12,16,20)
```

a-)

```
errors <- rnorm(n=5, mean=0, sd=5)
dat <- data.frame(Y = 20 + 4*X + errors, X = X, e = errors)
Xh <- 10

fit_p2 <- lm(Y ~ X, data = dat)
b0_p2 <- fit_p2$coefficients[1]
b1_p2 <- fit_p2$coefficients[2]
Yh <- b0_p2 + b1_p2 * Xh

confint <- predict(fit_p2, data.frame(X=Xh), interval = "confidence", level = 0.95)
```

$b_0 = 18.399$
$b_1 = 4.194$

Y_h (at $X_h = 10$) $= 18.399 + 4.194 * 10 = 60.3389$

95% confidence interval: $54.272 \leq E(Y_h) \leq 66.406$

b-)

```
p2 <- replicate(200, {
  errors <- rnorm(n=5, mean=0, sd=5)
  dat_200 <- data.frame(Y = 20 + 4*X + errors, X = X, e = errors)
  fit_200 <- lm(Y ~ X, data = dat_200)
  b0_200 <- fit_200$coefficients[1]
  b1_200 <- fit_200$coefficients[2]
  conf <- predict(fit_200, data.frame(X=10), interval = "confidence", level = 0.95)

  out <- c(b0_200, b1_200, conf[1], conf[2], conf[3])
  out
})
p2 <- t(p2)
p2 <- data.frame(p2)
```

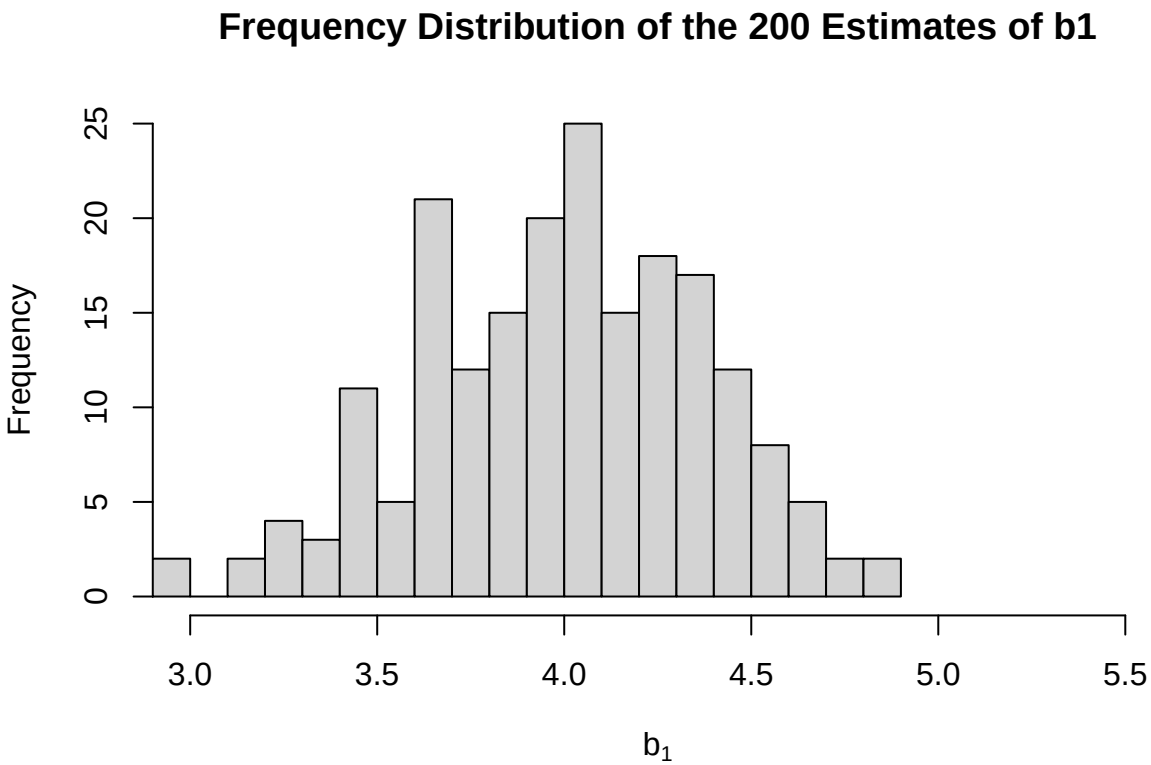
```r
colnames(p2) <- c("b0","b1","Yh","lower","upper")
head(p2)
```

```
##         b0       b1       Yh    lower    upper
## 1 24.65440 3.494778 59.60218 53.28720 65.91716
## 2 16.95306 4.404395 60.99701 54.08623 67.90780
## 3 18.81925 4.254555 61.36480 55.77090 66.95869
## 4 21.61641 3.844115 60.05756 52.88830 67.22682
## 5 17.89215 4.202895 59.92109 46.82590 73.01629
## 6 21.76661 3.489880 56.66541 48.72731 64.60351
```

The results for 200 coefficients, Y_h, and lower/upper bounds of confidence intervals are stored in the dataframe called 'p2', and its first six rows are shown above.

c-)

```r
hist(p2$b1, breaks = 20, xlim = c(3.0, 5.5), main = "Frequency Distribution of the 200 Estimates of b1"
```




```r
mean_b1 <- mean(p2[,2])
sd_b1 <- sqrt(var(p2[,2]))

# Theoretical Expectations
exp_b1mean <- 4 # coefficient of X in true regression
exp_b1sd <- sqrt(25 / sum((X - mean(X))^2)) # 25 is given in the problem as variance of E(Y)
```

Mean of 200 estimates of $b_1 = 3.9946$

SD of 200 estimates of $b_1 = 0.3906$

Theoretical expectation of mean of $b_1 = 4$

Theoretical expectation of SD of $b_1 = 0.3953$

For both mean and standard deviation, the estimates and the theoretical expectations are consistent to each other as the numbers show above.

d-)

```
mean_Yh <- mean(p2[,3])
notYh <- p2 %>% filter(lower > mean_Yh | upper < mean_Yh) %>% count()
propor <- notYh / nrow(p2)
```

6% of the 200 confidence intervals for $E(Y_h)$ when $X_h = 10$ include $E(Y_h)$, and this result is consistent with theoretical expectations as we had constructed 95% confidence intervals.

Problem 3

```
question3 <- read.csv("/cloud/project/Data/Question3.csv")
```

a-)

```
set.seed(1023)
train_index <- sort(sample(1:nrow(question3), 400))
train_set <- question3[train_index,]
test_set <- question3[-train_index,]
```

b-)

```
N <- nrow(train_set)
fit_q3 <- lm(Y ~ X, data = train_set)
b0_q3 <- fit_q3$coefficients[1]
b1_q3 <- fit_q3$coefficients[2]
summary(fit_q3)
```

```
##
## Call:
## lm(formula = Y ~ X, data = train_set)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -226047  -36227   -5112   25608  397092
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept) -74973.827  12077.290  -6.208 1.35e-09 ***
## X              155.012      5.104  30.373  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 73410 on 398 degrees of freedom
## Multiple R-squared:  0.6986, Adjusted R-squared:  0.6978
## F-statistic: 922.5 on 1 and 398 DF,  p-value: < 2.2e-16
```

```
f2 <- qf(0.95,1,N-2)
```

$\mathrm{Y} = -7.4973827 \times 10^4 + 155.0122\mathrm{X}$

Above regression model is significant as the summary shows that F-statistic (922.5) is much bigger than $F(1-\alpha; 1, n-2) = 3.86$, and p-value is very small.

c-)

```r
anova(fit_q3)
```

```
## Analysis of Variance Table
##
## Response: Y
##            Df     Sum Sq    Mean Sq F value    Pr(>F)
## X           1 4.9714e+12 4.9714e+12  922.52 < 2.2e-16 ***
## Residuals 398 2.1448e+12 5.3889e+09
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
SSE.F <- anova(fit_q3)$`Sum Sq`[2]
f_q3 <- anova(fit_q3)$`Mean Sq`[1] / anova(fit_q3)$`Mean Sq`[2]
```

d-)

```r
fit_q3red <- lm(Y ~ 1, data = train_set)
b0_q3red <- fit_q3red$coefficients[1]
anova(fit_q3red)
```

```
## Analysis of Variance Table
##
## Response: Y
##            Df     Sum Sq    Mean Sq F value Pr(>F)
## Residuals 399 7.1162e+12 1.7835e+10
```

```r
SSE.R <- anova(fit_q3red)$`Sum Sq`[1]
```

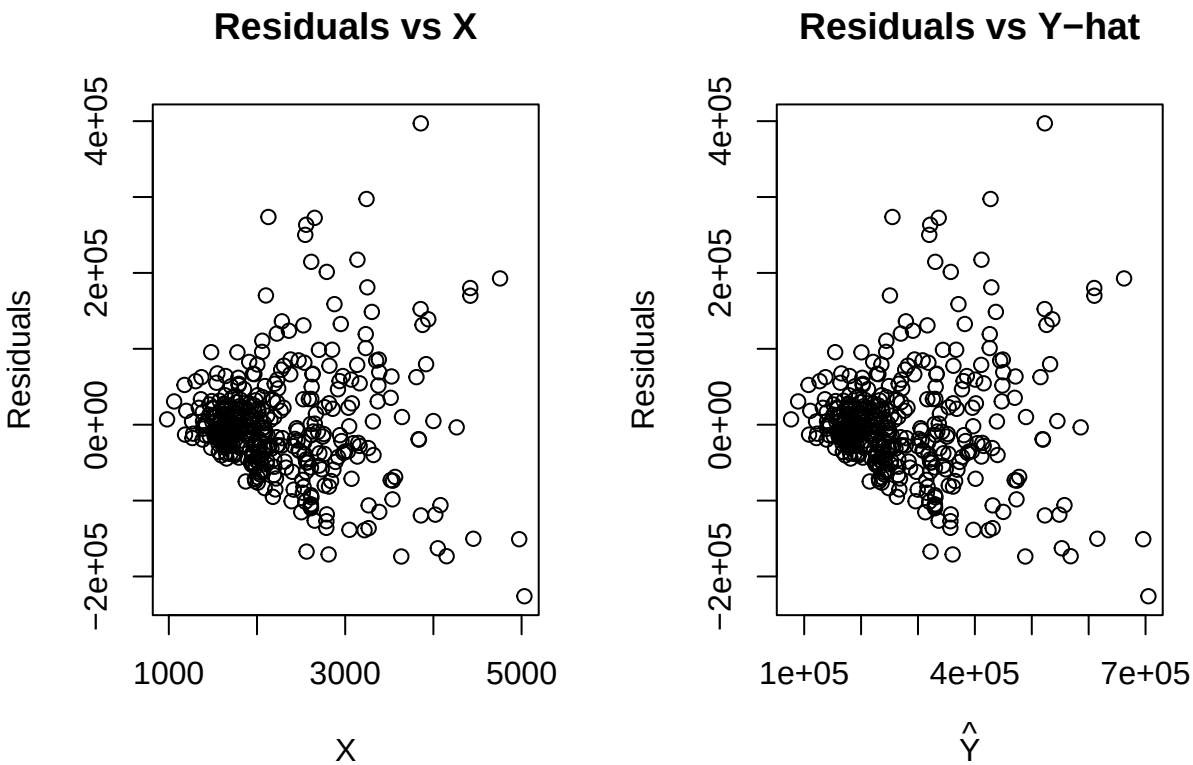
Full Model: $Y_i = \beta_0 + \beta_1 X_i + \epsilon_i = -7.4973827 \times 10^4 + 155.0122 X_i + \epsilon_i$

Reduced Model: $Y_i = \beta_0 + \epsilon_i = 2.7449955 \times 10^5 + \epsilon_i$

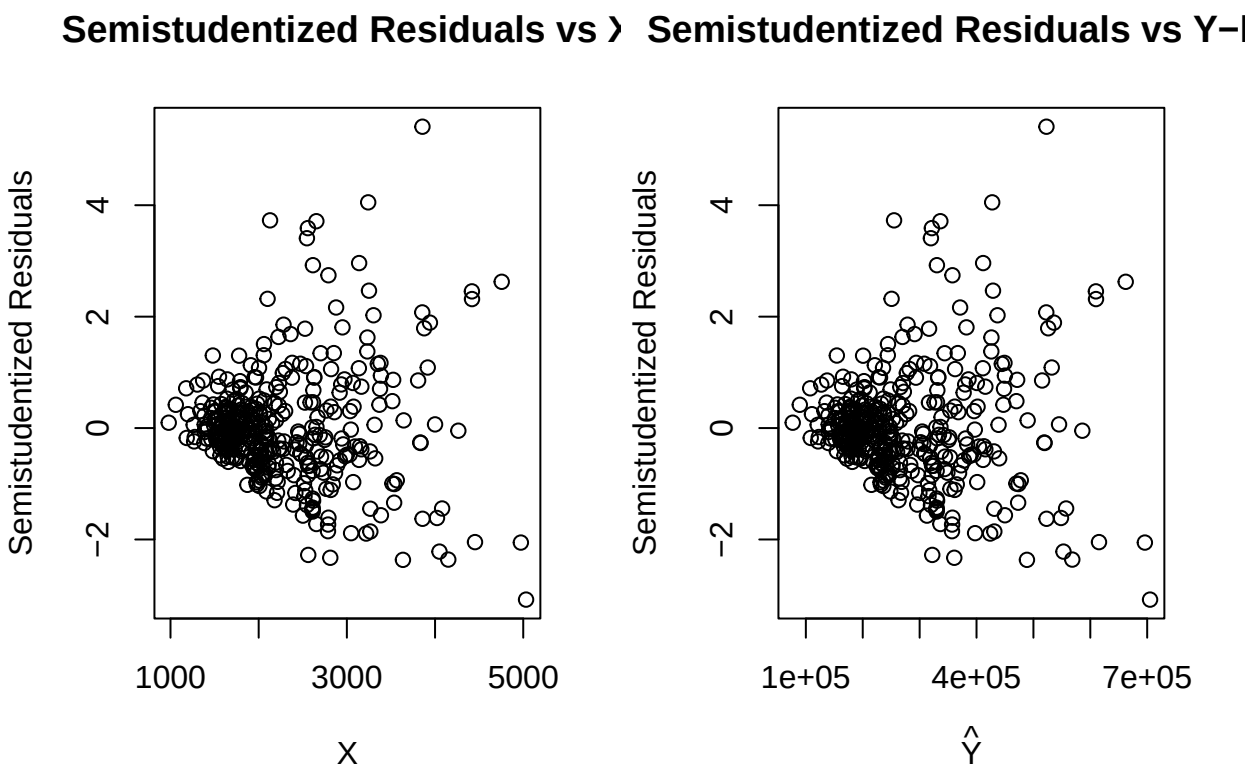
ANOVA tables for the full and reduced model show that SSE(R) $= 7.1161537 \times 10^{12}$ drops significantly to SSE(F) $= 2.1447781 \times 10^{12}$ when X is introduced to the full model. It suggests that H_a holds, and $F^* = \frac{MSR}{MSE}$ =922.52> $F(1-\alpha; 1, n-2)$ =3.86, so we conclude H_a.

e-)

```r
par(mfrow=c(1,2))
y_hat <- fit_q3$fitted.values
resid_q3 <- fit_q3$residuals
plot(train_set$X, resid_q3, xlab = "X", ylab = "Residuals", main = "Residuals vs X")
plot(y_hat, resid_q3, xlab = expression(hat(Y)), ylab = "Residuals", main = "Residuals vs Y-hat")
```




```
par(mfrow=c(1,2))
mse <- anova(fit_q3)$`Mean Sq`[2]
semi_residual <- resid_q3 / sqrt(mse)
plot(train_set$X, semi_residual, xlab = "X", ylab = "Semistudentized Residuals", main = "Semistudentize
plot(y_hat, semi_residual, xlab = expression(hat(Y)), ylab = "Semistudentized Residuals", main = "Semis
```



The normal residual plots show unequal variances as the dispersion of data points gets wider as X or $\hat{Y}$ increase.

The semistudentized residual plots show that some points are about 4 or more standard deviations away, which indicates that they could be outliers.

f-)

```
train_q3 <- data.frame(obs = train_set$Y, pred = fit_q3$fitted.values)
defaultSummary(train_q3)
```

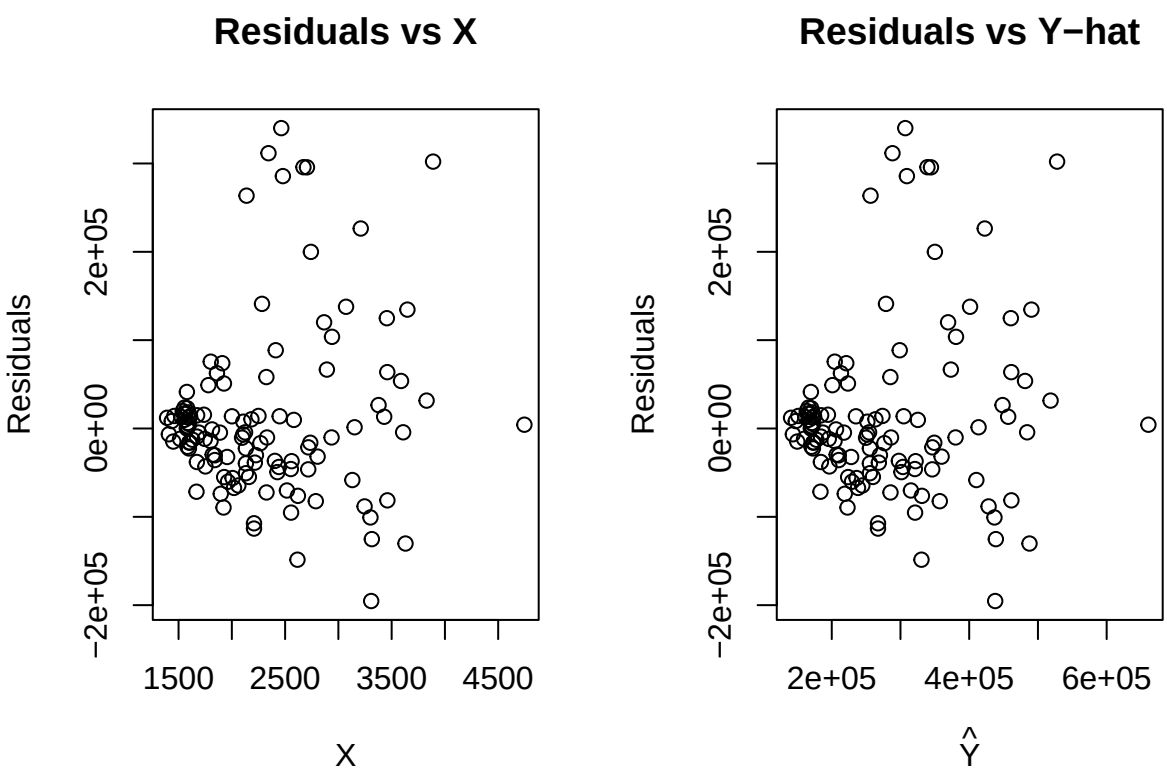
```
##         RMSE     Rsquared          MAE
## 7.322530e+04 6.986043e-01 4.977223e+04
```

```
predicted <- predict(fit_q3, test_set)
test_q3 <- data.frame(obs = test_set$Y, pred = predicted)
defaultSummary(test_q3)
```

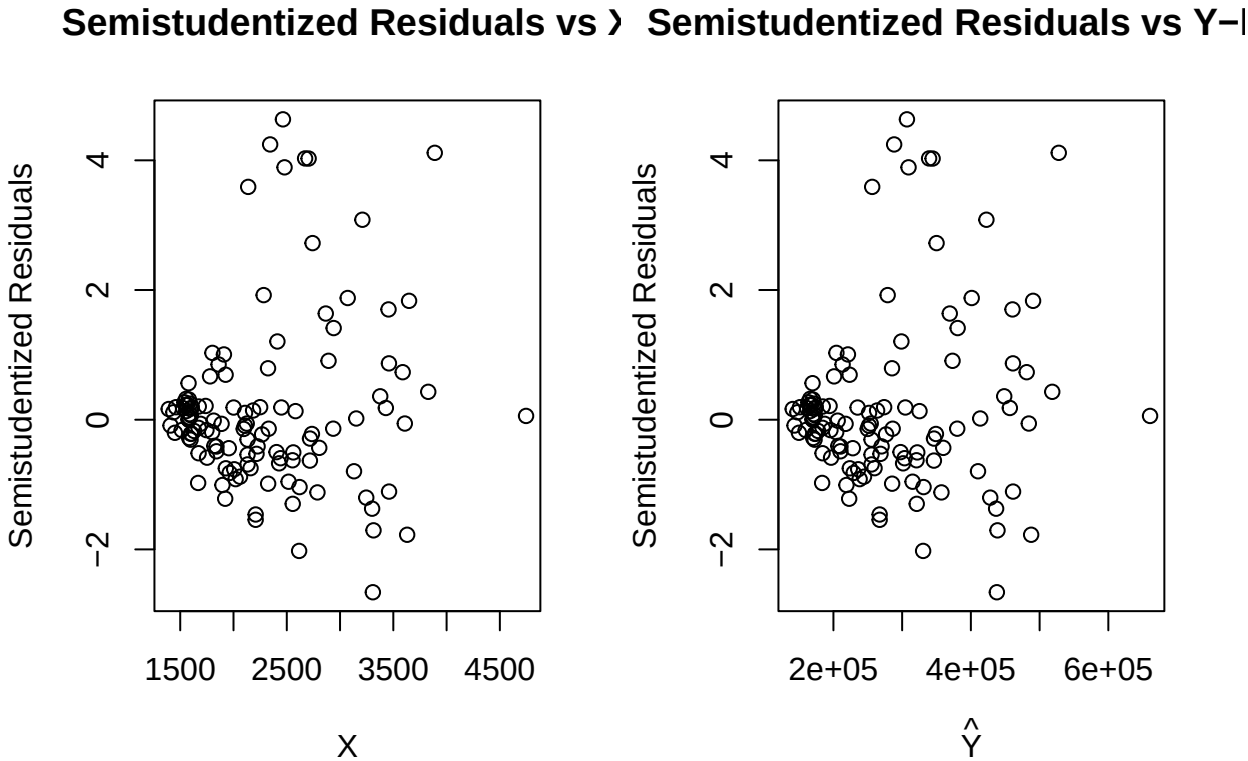
```
##         RMSE     Rsquared          MAE
## 9.572198e+04 6.085138e-01 6.103652e+04
```

g-)

```
par(mfrow=c(1,2))
resid_test <- test_q3$obs - test_q3$pred
plot(test_set$X, resid_test, xlab = "X", ylab = "Residuals", main = "Residuals vs X")
plot(test_q3$pred, resid_test, xlab = expression(hat(Y)), ylab = "Residuals", main = "Residuals vs Y-hat
```




```
par(mfrow=c(1,2))
semi_residual2 <- resid_test / sqrt(mse)
plot(test_set$X, semi_residual2, xlab = "X", ylab = "Semistudentized Residuals", main = "Semistudentize
plot(test_q3$pred, semi_residual2, xlab = expression(hat(Y)), ylab = "Semistudentized Residuals", main =
```



As it was shown the same in the training data, the test dataset also show that the variances increase with X and $\hat{Y}$, and the semistudentized residuals show that some data points are 4 standard deviations away, indicating possible outliers.

h-)

Although we need to be careful, the model suggests that we should consider removing outliers in the dataset. If we finally confirm the existence of outliers after further analysis, we can remove them and will be able to build a better model. The current model seems robust because RMSE, R-Squared, and MAE values are consistent between training and test datasets. Also, the R-Squared value slightly below 0.7 indicates that it is a fine model, but there is a possibility to improve it by employing other predictor variables.